

EEL 6814

HMW 5

Due November 7, 2024

In this homework you will experiment with transfer learning. You will find in the course website a trained AlexNet for the EMNIST letter split (<https://www.nist.gov/itl/products-and-services/emnist-dataset>). The EMNIST letter split contains 124,800 grayscale training images of 26 handwritten English letters.

The goal of the homework is to transfer the information contained in its parameters to classify handwritten digits. A subset of the digits data set is also in the website and it is small, just 1,000 digits (samples). Train the best you can a small CNN (also provided in the website with random weights) with 700 digits (500 training/ 200 for cross validation), and 300 digits for testing. Explain how you select the hyper parameters and show the performance in a confusion matrix.

Improve this baseline performance of the CNN by using the well-trained AlexNet, freezing its weights, and stripping the classification layers. First. Use the digits as inputs to the AlexNet and train with their targets. Please implement the two basic methodologies for fine tuning we discussed in the lecture: the depth augmented and the width augmented architectures. Pay particular attention to the scaling and normalization, i.e. cross validate it with the 200 crossvalidation digits. Use the 700 digits training set to fine tune the extra layer and the classifier. You can experiment with one or two layers classifier. Test the performance of the AlexNet with transfer learning in the 300 test digits and present the confusion matrix.

Compare the AlexNet performance with the original CNN trained for the task in the small 1,000 digits. Discuss the results you obtained.